ADVANCE TRAUMA

ATLS® training compared with
standard care in adult trauma patients

A batched stepped-wedge cluster randomised trial in India
NCT06321419 · CTRI/2024/07/071336

Background and rationale

Around 5 million people die from injuries each year[1]

ATLS® is the most widely used trauma life support programme, taught in more than 80 countries[2]

Observational studies suggest better survival, but high-quality evidence on mortality is lacking[3]



1. GBD 2023 Causes of Death Collaborators. Lancet. 2025;406:1811–1872.
2. American College of Surgeons. Advanced Trauma Life Support®.
3. Nakhid Z et al. Scand J Trauma Resusc Emerg Med. 2026;34:15.

Purpose and content

> The ATLS Course emphasizes rapid assessment and concurrent treatment of severely injured patients. Care starts at the time of injury and continues through initial assessment, lifesaving interventions, reevaluation, stabilization, and, when needed, transfer.

American College of Surgeons. Advanced Trauma Life Support® (ATLS®) Student Course Manual. 11th ed. 2025.

Impact on clinicians

> There is abundant evidence that ATLS training improves knowledge base, psychomotor skills, application of skills in resuscitation, and the confidence and performance of clinicians. The organizational and procedural skills taught in the course are retained by course participants for at least 6 years, which may be the most significant impact.

American College of Surgeons. Advanced Trauma Life Support® (ATLS®) Student Course Manual. 11th ed. 2025.

Impact on patients

> ATLS training in a developing country has resulted in a decrease in injury mortality. Lower-per-capita rates of deaths from injuries are observed in areas where clinicians have ATLS training. In one study, a small trauma care team led by a doctor with ATLS experience had equivalent patient survival when compared with a larger team with more doctors in an urban setting. In addition, there were more unexpected survivors than fatalities.

American College of Surgeons. Advanced Trauma Life Support® (ATLS®) Student Course Manual. 11th ed. 2025.

What have systematic reviews found?

Educational benefit; uncertain survival benefit

- Mohammad et al. (2014): 23 studies; knowledge and skills improve, but strong mortality evidence is lacking[1]
- Cochrane (2014): no eligible controlled trials of ATLS effects on mortality or morbidity[2]

Observational mortality evidence

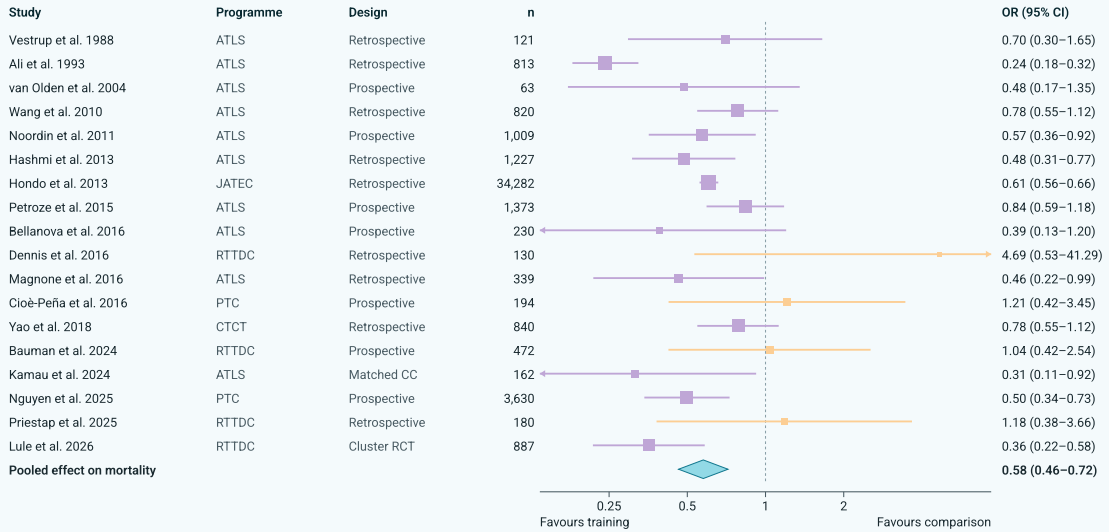
- Putra et al. (2023): seven studies; OR 0.68 (95% CI 0.39–1.20)[3]
- Nakhid et al. (2026): 17 observational studies of trauma life support training; OR 0.60 (95% CI 0.48–0.75)[4]

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1. World J Surg. 2014;38:322–329. 2. Cochrane Database Syst Rev. 2014;CD004173.
3. New Ropanasuri J Surg. 2023;8(2). 4. Scand J Trauma Resusc Emerg Med. 2026;34:15.

Updated systematic review

▸ Play timeline | All studies | RCT | Prospective | Retrospective | ATLS | PTC | RTTDC | JATEC | CTCT



Random-effects OR 0.58 (95% CI 0.46–0.72); I² 75%; 18 studies

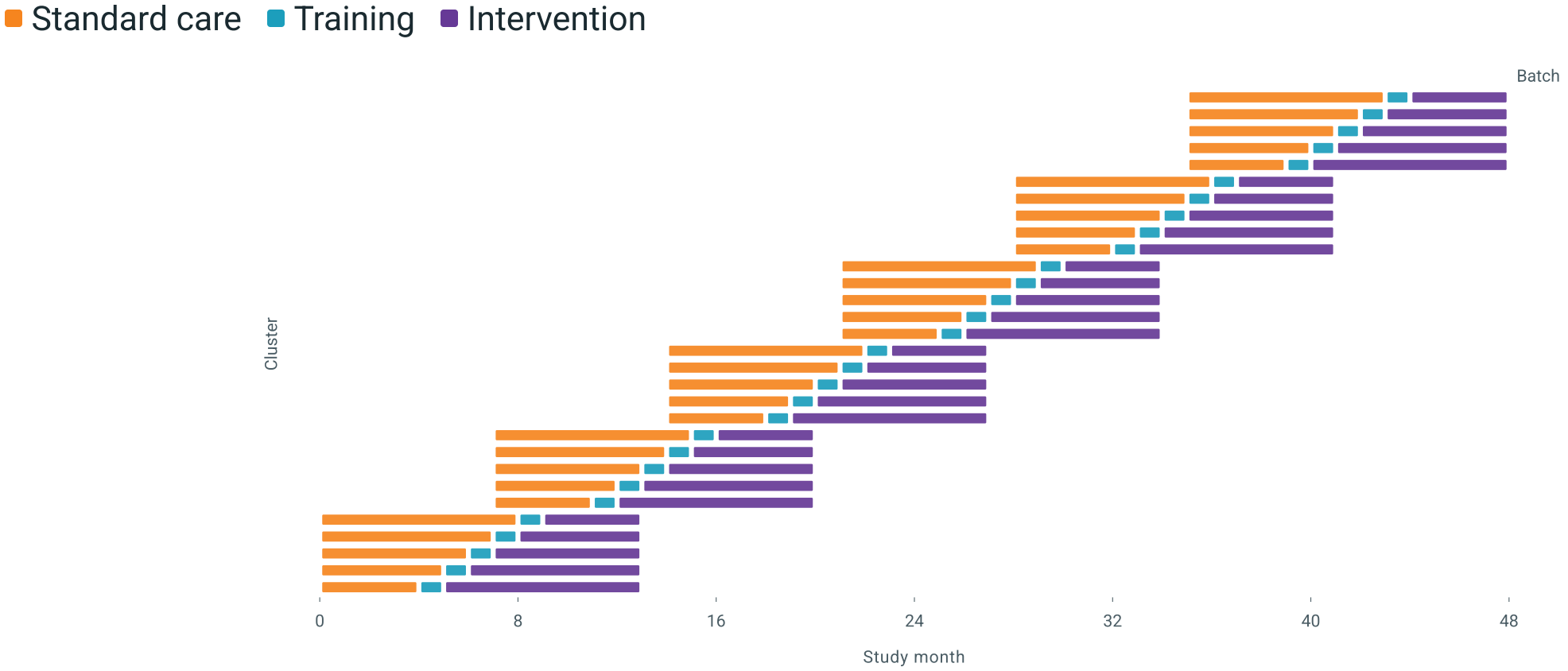
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Nakhid et al. Scand J Trauma Resusc Emerg Med. 2026;34:15, updated with Lule et al. (2026). Multiple training programmes; Lule evaluated RTTDC, not ATLS®.

Aim

To compare the effects of ATLS® training with standard care on outcomes in adult trauma patients.

Batched stepped-wedge cluster randomised trial



Setting and clusters

Hospitals in India

- 30 secondary or tertiary hospitals providing trauma care
- Round-the-clock surgical and orthopaedic emergency services
- No more than 25% of initial-trauma-care physicians formally trained in trauma life support

Participating units

- Usually one or two units per hospital
- At least 12 trauma admissions or transfers for admission per month
- Patient data collected on participating units’ working days

Patient eligibility

All inclusion criteria

- Aged 15 years or older; injury less than 48 hours before arrival
- Seen in the ED for trauma (ICD-10 chapter XX)
- Admitted, died between arrival and admission, or transferred from the ED for admission
- Managed by a participating ED unit

Exclusions

- Isolated closed extremity fracture
- Direct ward admission without assessment by an ED physician

Intervention and comparator

Standard care



Usual initial trauma resuscitation by casualty medical officers or residents, before trial-provided ATLS training.

ATLS® training

Randomisation and masking



Hospital-level allocation

- Six batches of five hospitals; five implementation sequences

A 2.5-day course at an accredited facility in India. Around ten physicians per hospital, usually from one or two units.

- One hospital per sequence in each batch
- Covariate-constrained randomisation balances monthly eligible-patient volume

Concealment and masking

- Training order concealed as long as logistically possible
- Advance notification allows course arrangements
- Blinding is not possible in this pragmatic training trial

What happens at each hospital?

13 months of inclusion

- At least four months of standard care
- One-month transition phase for ATLS training
- At least four months after training; timing depends on randomised sequence

Follow-up and nested study

- Each patient followed for 90 days from ED arrival
- Nested measurements in the three months before and three months after transition
- Transition-phase data excluded from the main analysis

Primary outcome

In-hospital mortality within 30 days of emergency department arrival.

Death during the initial hospital admission; distinct from all-cause 30-day mortality.

Secondary outcomes

Main trial

- All-cause mortality at 24 hours, 30 days and 90 days
- ED, ICU and hospital length of stay; ICU admission
- Return to work at 30 and 90 days

Nested staircase study

- Adherence to ATLS principles during initial care
- Health-related quality of life: EQ-5D-5L
- Disability: WHODAS 2.0

Schedule of patient assessments

Assessment	During admission	After discharge	Day 30	Day 90
Eligibility, injury and clinical characteristics	ED / admission	—	—	—
In-hospital primary outcome	Records through day 30 of initial admission	—	—	—
All-cause mortality	24 hours	—	Yes	Yes
Return to work	—	—	Yes	Yes
ATLS adherence · nested	First hour of physician care	—	—	—
EQ-5D-5L / WHODAS · nested	—	Within 7 days	Yes	Yes

Time points are relative to ED arrival unless stated. EQ-5D-5L and WHODAS are assessed in survivors. Lengths of stay and ICU admission come from records.

Sample size

At least 4,320 patients for analysis

- 30 hospitals × 12 patients per month × 12 non-transition months
- Approximately 90% power for mortality of 20% versus 15%
- Five percentage-point absolute reduction; 25% relative reduction

Allowing for clustering

- Assumed intracluster correlation: 0.02
- Cluster autocorrelation: 0.9; time-decay structure
- Variable cluster sizes considered in power calculations

At the same recruitment rate across all 13 months: about 156 inclusions per hospital, of which 144 are outside transition.

Statistical analysis

Compare as scheduled

- Analyse available observations regardless of training receipt
- Assign exposure by the scheduled transition date
- Exclude the one-month transition phase

Account for design and time

- Mixed-effects models adjust for study month and hospital / hospital-period clustering
- Report odds ratios and, where models converge, risk differences with 95% confidence intervals
- Two-sided 5% significance level; secondary outcomes interpreted without multiplicity adjustment

Data collection and quality

At the hospital

- Research coordinators collect records and clinical observations
- Follow up after discharge or transfer by telephone
- Follow-up telephone numbers retained locally

Central safeguards

- Secure electronic case report form with VPN and two-factor authentication
- Central checks, statistical monitoring and selective site visits
- Source verification and at least ten years of archiving

Ethics and consent for data collection

Opt out

- Routine records and observed adherence to ATLS principles
- Written information for patients or representatives
- May opt out before final analysis without affecting care

Opt in

- Consent for non-routinely collected data
- Ages 15–18: assent plus guardian or representative consent

Waiver

- When unable to consent and no legally authorised representative is available
- Seek consent when the patient or representative can provide it

Ethical review and site permissions are required before participation. Consent provisions concern data collection, not allocation of physician training.

Safety monitoring

Events tracked in both phases

- Mechanical ventilation for more than seven days
- New renal replacement therapy
- Vasopressors for more than two days, or restarted after at least two days off
- Other unexpected serious complications causing life threat, prolonged admission or meaningful harm

Trial oversight and dissemination

Reporting and review

- Report safety events to trial management within 24 hours
- Trial management assesses relatedness; local investigators follow events
- Probably trial-related events escalated immediately to the oversight committee

Governance

- Trial Team: daily operations
- Trial Management Group: clinical and practical conduct
- Joint Steering and Data Monitoring Committee: independent majority and independent chair
- Interim review focuses on safety and feasibility; no planned efficacy stopping

Transparency

- Protocol-listed funding: Swedish Research Council and Laerdal Foundation
- Publish findings and make metadata and anonymised data available as planned in the protocol

What will this trial tell us?

Patient benefit

- Does introducing ATLS training improve survival in this setting?
- Do changes in initial care translate into better recovery?

Policy and practice

- Inform decisions about investing in trauma life support training
- Interpret effects alongside uncertainty, implementation and the participating hospitals' context
- Avoid assuming that a non-significant result proves no benefit

Questions?

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